RNA-Seq analysis mini-project

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##Section 1. Differential Expression Analysis

```r
library(DESeq2)
```

```
Warning: package 'DESeq2' was built under R version 4.5.2

Loading required package: S4Vectors

Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics


Attaching package: 'generics'

The following objects are masked from 'package:base':

    as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
    setequal, union


Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

    IQR, mad, sd, var, xtabs
```

```
The following objects are masked from 'package:base':

    anyDuplicated, aperm, append, as.data.frame, basename, cbind,
    colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
    get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
    order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
    rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
    unsplit, which.max, which.min


Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

    findMatches

The following objects are masked from 'package:base':

    expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomicRanges

Warning: package 'GenomicRanges' was built under R version 4.5.2

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Warning: package 'SummarizedExperiment' was built under R version 4.5.2

Loading required package: MatrixGenerics

Warning: package 'MatrixGenerics' was built under R version 4.5.2

Loading required package: matrixStats
```

```
Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

    colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
    colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
    colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
    colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
    colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
    colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
    colWeightedMeans, colWeightedMedians, colWeightedSds,
    colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
    rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
    rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
    rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
    rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
    rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
    rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
    rowWeightedSds, rowWeightedVars

Loading required package: Biobase

Warning: package 'Biobase' was built under R version 4.5.2

Welcome to Bioconductor

    Vignettes contain introductory material; view with
    'browseVignettes()'. To cite Bioconductor, see
    'citation("Biobase")', and for packages 'citation("pkgname")'.


Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

    rowMedians

The following objects are masked from 'package:matrixStats':

    anyMissing, rowMedians
```

```r
metaFile  <- "GSE37704_metadata.csv"
countFile <- "GSE37704_featurecounts.csv"
```

```r
metaFile  <- "GSE37704_metadata (1).csv"
countFile <- "GSE37704_featurecounts.csv"

colData  <- read.csv(metaFile, row.names = 1)
countData <- read.csv(countFile, row.names = 1)
```

Q. Complete the code below to remove the troublesome first column from countData

```r
countData <- as.matrix(countData[, -1])
head(countData)
```

```
                SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000186092         0         0         0         0         0         0
ENSG00000279928         0         0         0         0         0         0
ENSG00000279457        23        28        29        29        28        46
ENSG00000278566         0         0         0         0         0         0
ENSG00000273547         0         0         0         0         0         0
ENSG00000187634       124       123       205       207       212       258
```

Q. Complete the code below to filter countData to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns).

```r
countData <- countData[rowSums(countData) > 0, ]
head(countData)
```

```
                SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000279457        23        28        29        29        28        46
ENSG00000187634       124       123       205       207       212       258
ENSG00000188976      1637      1831      2383      1226      1326      1504
ENSG00000187961       120       153       180       236       255       357
ENSG00000187583        24        48        65        44        48        64
ENSG00000187642         4         9        16        14        16        16
```

##Running DESeq2

```r
dds <- DESeqDataSetFromMatrix(
  countData = countData,
  colData   = colData,
  design    = ~ condition
)
```

```
Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
design formula are characters, converting to factors
```

```r
dds <- DESeq(dds)
```

```
estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing
```

```r
dds
```

```
class: DESeqDataSet
dim: 15975 6
metadata(1): version
assays(4): counts mu H cooks
rownames(15975): ENSG00000279457 ENSG00000187634 ... ENSG00000276345
  ENSG00000271254
rowData names(22): baseMean baseVar ... deviance maxCooks
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371
colData names(2): condition sizeFactor
```

```r
res <- results(dds)
res
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 15975 rows and 6 columns
                 baseMean log2FoldChange     lfcSE       stat      pvalue
                <numeric>      <numeric> <numeric>  <numeric>   <numeric>
ENSG00000279457   29.9136      0.1792571 0.3248215   0.551863 5.81042e-01
ENSG00000187634  183.2296      0.4264571 0.1402658   3.040350 2.36304e-03
ENSG00000188976 1651.1881     -0.6927205 0.0548465 -12.630156 1.43993e-36
ENSG00000187961  209.6379      0.7297556 0.1318599   5.534326 3.12428e-08
ENSG00000187583   47.2551      0.0405765 0.2718928   0.149237 8.81366e-01
...                   ...            ...       ...        ...         ...
ENSG00000273748  35.30265       0.674387  0.303666   2.220817 2.63633e-02
ENSG00000278817   2.42302      -0.388988  1.130394  -0.344118 7.30758e-01
ENSG00000278384   1.10180       0.332991  1.660261   0.200565 8.41039e-01
ENSG00000276345  73.64496      -0.356181  0.207716  -1.714752 8.63908e-02
ENSG00000271254 181.59590      -0.609667  0.141320  -4.314071 1.60276e-05
                      padj
                 <numeric>
ENSG00000279457 6.86555e-01
ENSG00000187634 5.15718e-03
ENSG00000188976 1.76553e-35
ENSG00000187961 1.13413e-07
ENSG00000187583 9.19031e-01
...                     ...
ENSG00000273748 4.79091e-02
ENSG00000278817 8.09772e-01
ENSG00000278384 8.92654e-01
ENSG00000276345 1.39761e-01
ENSG00000271254 4.53647e-05
```

Q. Call the summary() function on your results to get a sense of how many genes are up or down-regulated at the default 0.1 p-value cutoff.

```r
summary(res)
```

```
out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)       : 4349, 27%
LFC < 0 (down)     : 4396, 28%
outliers [1]       : 0, 0%
low counts [2]     : 1237, 7.7%
```

```
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

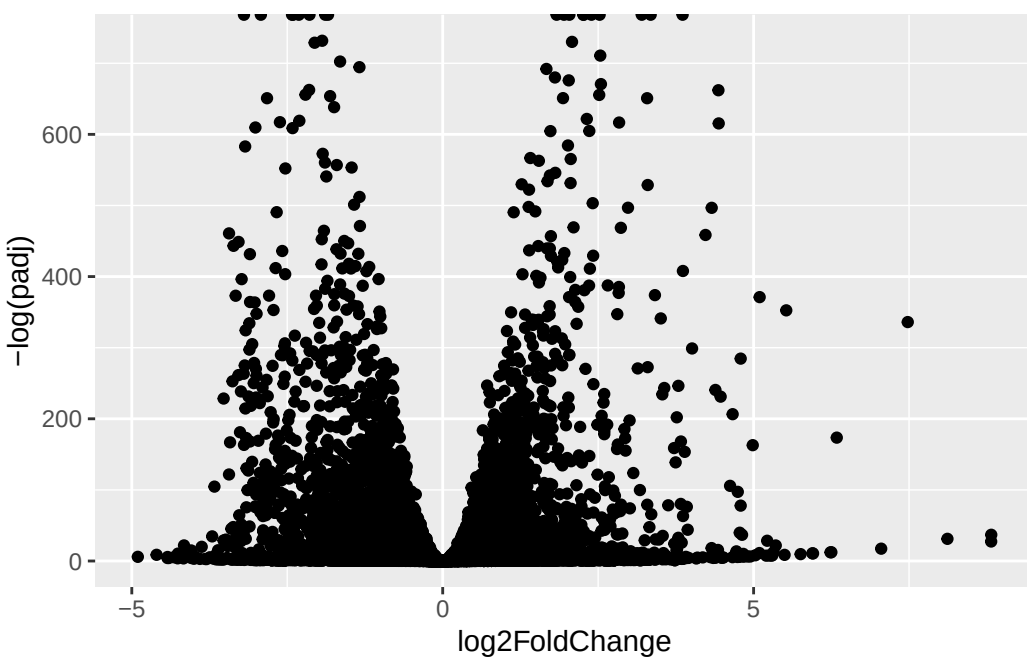
##Volcano Plot

```r
library(ggplot2)
```

```
Warning: package 'ggplot2' was built under R version 4.5.2
```

```r
ggplot(res) +
  aes(x = log2FoldChange, y = -log(padj)) +
  geom_point()
```

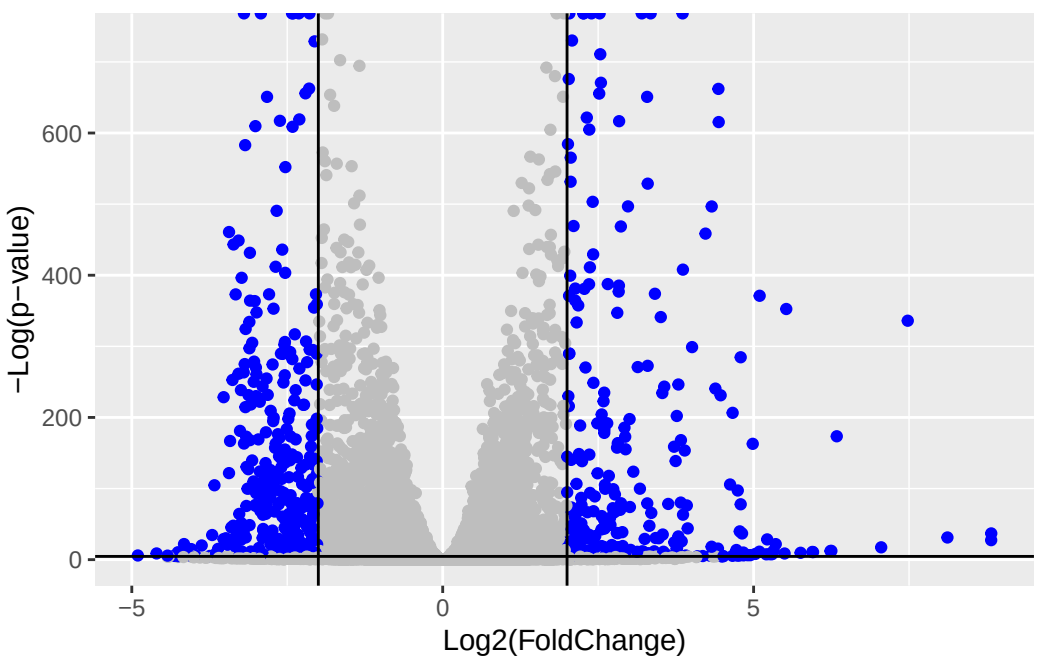
```
Warning: Removed 1237 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Q. Improve this plot by completing the below code, which adds color, axis labels and cutoff lines:

```r
library(ggplot2)

# Make a color vector for all genes
mycols <- rep("gray", nrow(res))

# Color blue the genes with fold change above 2
mycols[ abs(res$log2FoldChange) > 2 ] <- "blue"

# Color gray those with adjusted p-value more than 0.01
mycols[ res$padj > 0.01 ] <- "gray"

ggplot(res) +
  aes(x = log2FoldChange, y = -log(padj)) +
  geom_point( color= mycols) +
  xlab("Log2(FoldChange)") +
  ylab("-Log(p-value)") +
  geom_vline(xintercept = c(-2, 2)) +
  geom_hline(yintercept = -log(0.01))
```

```
Warning: Removed 1237 rows containing missing values or values outside the scale range
(`geom_point()`).
```



##Adding gene annotation

> Q. Use the mapIDs() function multiple times to add SYMBOL, ENTREZID and GENENAME annotation to our results by completing the code below.

```
library("AnnotationDbi")
```

```
Warning: package 'AnnotationDbi' was built under R version 4.5.2
```

```
library("org.Hs.eg.db")
```

```
columns(org.Hs.eg.db)
```

```
 [1] "ACCNUM"       "ALIAS"        "ENSEMBL"      "ENSEMBLPROT"  "ENSEMBLTRANS"
 [6] "ENTREZID"     "ENZYME"       "EVIDENCE"     "EVIDENCEALL"  "GENENAME"
[11] "GENETYPE"     "GO"           "GOALL"        "IPI"          "MAP"
[16] "OMIM"         "ONTOLOGY"     "ONTOLOGYALL"  "PATH"         "PFAM"
[21] "PMID"         "PROSITE"      "REFSEQ"       "SYMBOL"       "UCSCKG"
[26] "UNIPROT"
```

```
res$symbol <- mapIds(org.Hs.eg.db,
                     keys = row.names(res),
                     keytype = "ENSEMBL",
                     column = "SYMBOL",
                     multiVals = "first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```
res$entrez <- mapIds(org.Hs.eg.db,
                     keys = row.names(res),
                     keytype = "ENSEMBL",
                     column = "ENTREZID",
                     multiVals = "first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
res$name <- mapIds(org.Hs.eg.db,
                   keys = row.names(res),
                   keytype = "ENSEMBL",
                   column = "GENENAME",
                   multiVals = "first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
head(res, 10)
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 10 rows and 9 columns
                   baseMean log2FoldChange     lfcSE       stat      pvalue
                  <numeric>      <numeric> <numeric>  <numeric>   <numeric>
ENSG00000279457   29.913579      0.1792571 0.3248215   0.551863 5.81042e-01
ENSG00000187634  183.229650      0.4264571 0.1402658   3.040350 2.36304e-03
ENSG00000188976 1651.188076     -0.6927205 0.0548465 -12.630156 1.43993e-36
ENSG00000187961  209.637938      0.7297556 0.1318599   5.534326 3.12428e-08
ENSG00000187583   47.255123      0.0405765 0.2718928   0.149237 8.81366e-01
ENSG00000187642   11.979750      0.5428105 0.5215598   1.040744 2.97994e-01
ENSG00000188290  108.922128      2.0570638 0.1969053  10.446970 1.51281e-25
ENSG00000187608  350.716868      0.2573837 0.1027266   2.505522 1.22271e-02
ENSG00000188157 9128.439422      0.3899088 0.0467164   8.346302 7.04333e-17
ENSG00000237330    0.158192      0.7859552 4.0804729   0.192614 8.47261e-01
                     padj      symbol      entrez                   name
                  <numeric> <character> <character>            <character>
ENSG00000279457 6.86555e-01          NA          NA                     NA
ENSG00000187634 5.15718e-03      SAMD11      148398 sterile alpha motif ..
ENSG00000188976 1.76553e-35       NOC2L       26155 NOC2 like nucleolar ..
ENSG00000187961 1.13413e-07      KLHL17      339451 kelch like family me..
ENSG00000187583 9.19031e-01     PLEKHN1       84069 pleckstrin homology ..
ENSG00000187642 4.03379e-01       PERM1       84808 PPARGC1 and ESRR ind..
ENSG00000188290 1.30538e-24        HES4       57801 hes family bHLH tran..
ENSG00000187608 2.37452e-02       ISG15        9636 ISG15 ubiquitin like..
ENSG00000188157 4.21970e-16        AGRN      375790                  agrin
ENSG00000237330          NA      RNF223      401934 ring finger protein ..
```

Q. Finally for this section let's reorder these results by adjusted p-value and save them to a CSV file in your current project directory.

```r
res <- res[order(res$padj), ]
write.csv(res, file = "deseq_results.csv")
```

##Section 2. Pathway Analysis

##KEGG pathways

```r
library(pathview)
```

```
Warning: package 'pathview' was built under R version 4.5.2

##############################################################################
Pathview is an open source software package distributed under GNU General
Public License version 3 (GPLv3). Details of GPLv3 is available at
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
formally cite the original Pathview paper (not just mention it) in publications
or products. For details, do citation("pathview") within R.

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG
license agreement (details at http://www.kegg.jp/kegg/legal.html).
##############################################################################
```

```r
library(gage)
```

```
Warning: package 'gage' was built under R version 4.5.2
```

```r
library(gageData)

data(kegg.sets.hs)
data(sigmet.idx.hs)

kegg.sets.hs <- kegg.sets.hs[sigmet.idx.hs]

head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
[1] "10"   "1544" "1548" "1549" "1553" "7498" "9"

$`hsa00983 Drug metabolism - other enzymes`
 [1] "10"     "1066"   "10720"  "10941"  "151531" "1548"   "1549"   "1551"
 [9] "1553"   "1576"   "1577"   "1806"   "1807"   "1890"   "221223" "2990"
[17] "3251"   "3614"   "3615"   "3704"   "51733"  "54490"  "54575"  "54576"
[25] "54577"  "54578"  "54579"  "54600"  "54657"  "54658"  "54659"  "54963"
[33] "574537" "64816"  "7083"   "7084"   "7172"   "7363"   "7364"   "7365"
[41] "7366"   "7367"   "7371"   "7372"   "7378"   "7498"   "79799"  "83549"
[49] "8824"   "8833"   "9"      "978"

$`hsa00230 Purine metabolism`
  [1] "100"    "10201"  "10606"  "10621"  "10622"  "10623"  "107"    "10714"
  [9] "108"    "10846"  "109"    "111"    "11128"  "11164"  "112"    "113"
 [17] "114"    "115"    "122481" "122622" "124583" "132"    "158"    "159"
 [25] "1633"   "171568" "1716"   "196883" "203"    "204"    "205"    "221823"
 [33] "2272"   "22978"  "23649"  "246721" "25885"  "2618"   "26289"  "270"
 [41] "271"    "27115"  "272"    "2766"   "2977"   "2982"   "2983"   "2984"
 [49] "2986"   "2987"   "29922"  "3000"   "30833"  "30834"  "318"    "3251"
 [57] "353"    "3614"   "3615"   "3704"   "377841" "471"    "4830"   "4831"
 [65] "4832"   "4833"   "4860"   "4881"   "4882"   "4907"   "50484"  "50940"
 [73] "51082"  "51251"  "51292"  "5136"   "5137"   "5138"   "5139"   "5140"
 [81] "5141"   "5142"   "5143"   "5144"   "5145"   "5146"   "5147"   "5148"
 [89] "5149"   "5150"   "5151"   "5152"   "5153"   "5158"   "5167"   "5169"
 [97] "51728"  "5198"   "5236"   "5313"   "5315"   "53343"  "54107"  "5422"
[105] "5424"   "5425"   "5426"   "5427"   "5430"   "5431"   "5432"   "5433"
[113] "5434"   "5435"   "5436"   "5437"   "5438"   "5439"   "5440"   "5441"
[121] "5471"   "548644" "55276"  "5557"   "5558"   "55703"  "55811"  "55821"
[129] "5631"   "5634"   "56655"  "56953"  "56985"  "57804"  "58497"  "6240"
[137] "6241"   "64425"  "646625" "654364" "661"    "7498"   "8382"   "84172"
[145] "84265"  "84284"  "84618"  "8622"   "8654"   "87178"  "8833"   "9060"
[153] "9061"   "93034"  "953"    "9533"   "954"    "955"    "956"    "957"
[161] "9583"   "9615"
```

```r
foldchanges <- res$log2FoldChange
names(foldchanges) <- res$entrez

head(foldchanges)
```

```
     1266     54855      1465      2034      2150      6659
-2.422719  3.201955 -2.313738 -1.888019  3.344508  2.392288
```

```r
foldchanges <- foldchanges[!is.na(names(foldchanges))]
```

```r
# Get the results
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

```r
attributes(keggres)
```

```
$names
[1] "greater" "less"    "stats"
```

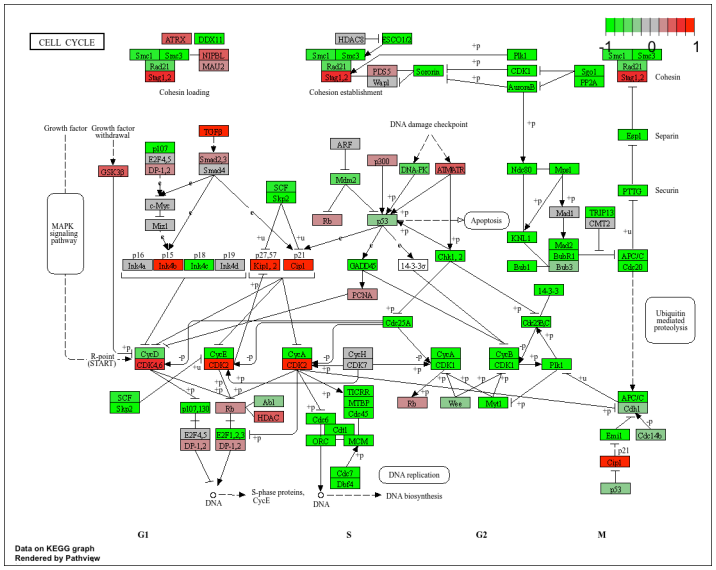
```r
# Look at the first few down (less) pathways
head(keggres$less)
```

```
                                          p.geomean stat.mean        p.val
hsa04110 Cell cycle                    9.366328e-06 -4.369134 9.366328e-06
hsa03030 DNA replication               9.592718e-05 -3.946505 9.592718e-05
hsa03013 RNA transport                 1.439613e-03 -3.014203 1.439613e-03
hsa03440 Homologous recombination      3.106306e-03 -2.848189 3.106306e-03
hsa04114 Oocyte meiosis                3.888649e-03 -2.688749 3.888649e-03
hsa00010 Glycolysis / Gluconeogenesis 9.125398e-03 -2.398384 9.125398e-03
                                            q.val set.size         exp1
hsa04110 Cell cycle                    0.001507979      121 9.366328e-06
hsa03030 DNA replication               0.007722138       36 9.592718e-05
hsa03013 RNA transport                 0.077259252      144 1.439613e-03
hsa03440 Homologous recombination      0.125028834       28 3.106306e-03
hsa04114 Oocyte meiosis                0.125214483      102 3.888649e-03
hsa00010 Glycolysis / Gluconeogenesis 0.213990959       53 9.125398e-03
```

```r
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/saharkamali/Desktop/BIMM 143 /Class 14/class14

Info: Writing image file hsa04110.pathview.png
```

```
knitr::include_graphics("hsa04110.pathview.png")
```




```
# A different PDF based output of the same data
pathview(gene.data=foldchanges, pathway.id="hsa04110", kegg.native=FALSE)
```

```
'select()' returned 1:1 mapping between keys and columns

Warning: reconcile groups sharing member nodes!

     [,1] [,2]
[1,] "9"  "300"
[2,] "9"  "306"

Info: Working in directory /Users/saharkamali/Desktop/BIMM 143 /Class 14/class14

Info: Writing image file hsa04110.pathview.pdf
```

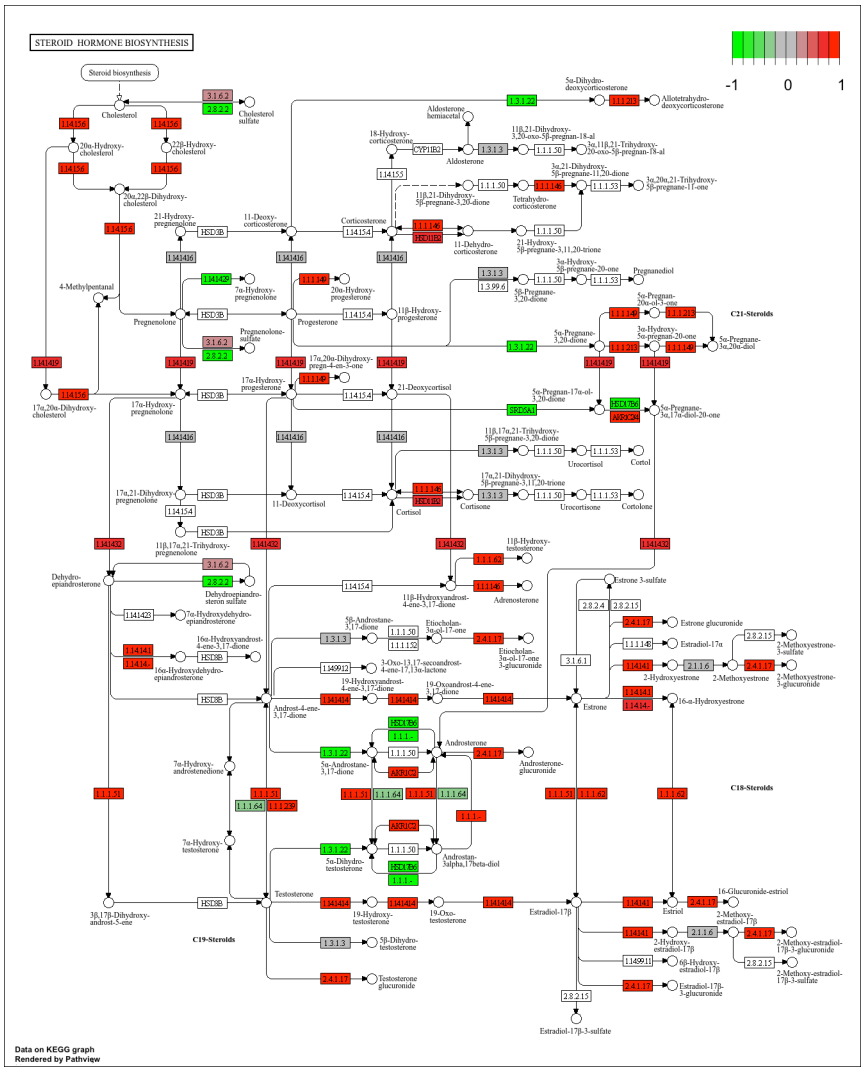
```
## Focus on top 5 upregulated pathways here for demo purposes only
keggrespathways <- rownames(keggres$greater)[1:5]

# Extract the 8 character long IDs part of each string
keggresids = substr(keggrespathways, start=1, stop=8)
keggresids
```

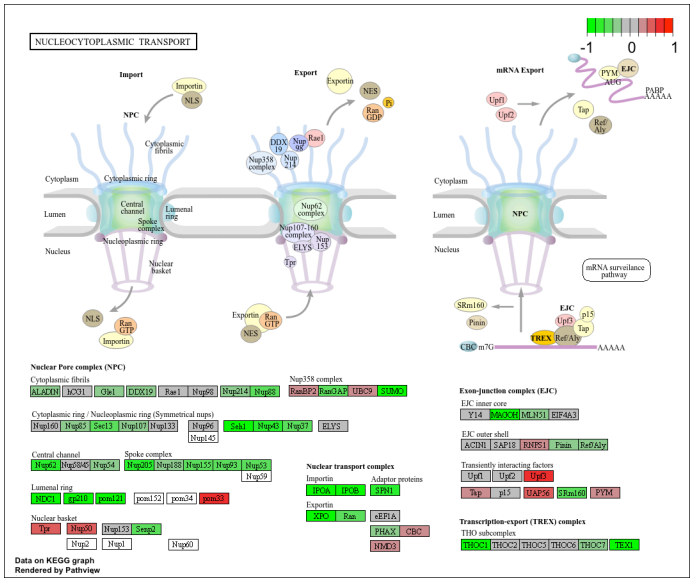
```
[1] "hsa04640" "hsa04630" "hsa00140" "hsa04142" "hsa04330"
```

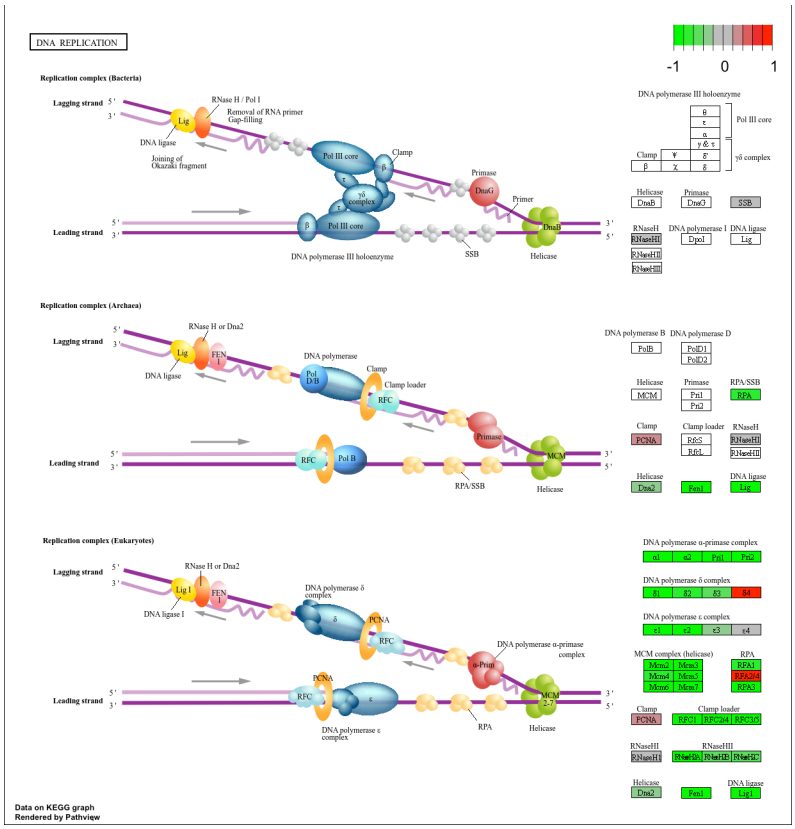
```r
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

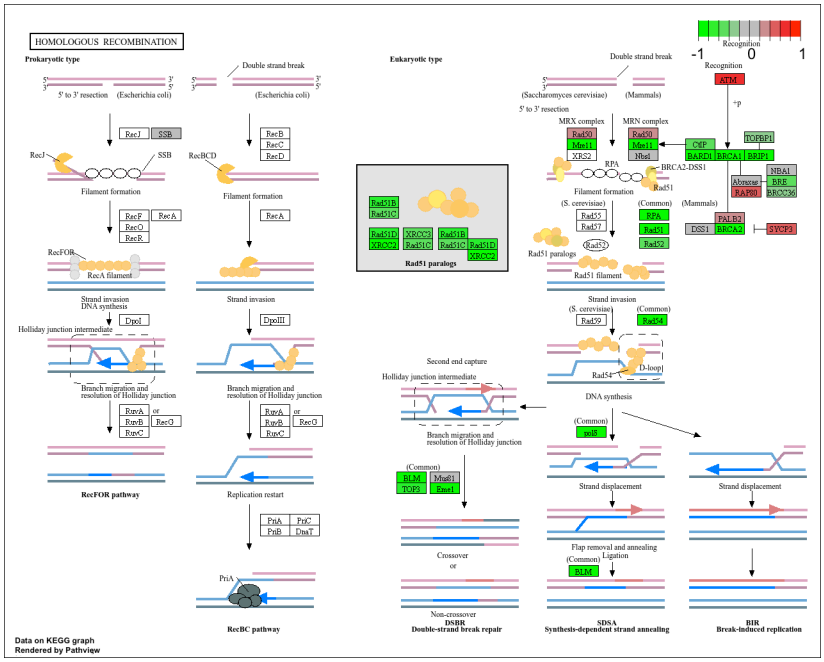
```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/saharkamali/Desktop/BIMM 143 /Class 14/class14

Info: Writing image file hsa04640.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/saharkamali/Desktop/BIMM 143 /Class 14/class14

Info: Writing image file hsa04630.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/saharkamali/Desktop/BIMM 143 /Class 14/class14

Info: Writing image file hsa00140.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/saharkamali/Desktop/BIMM 143 /Class 14/class14

Info: Writing image file hsa04142.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/saharkamali/Desktop/BIMM 143 /Class 14/class14

Info: Writing image file hsa04330.pathview.png
```

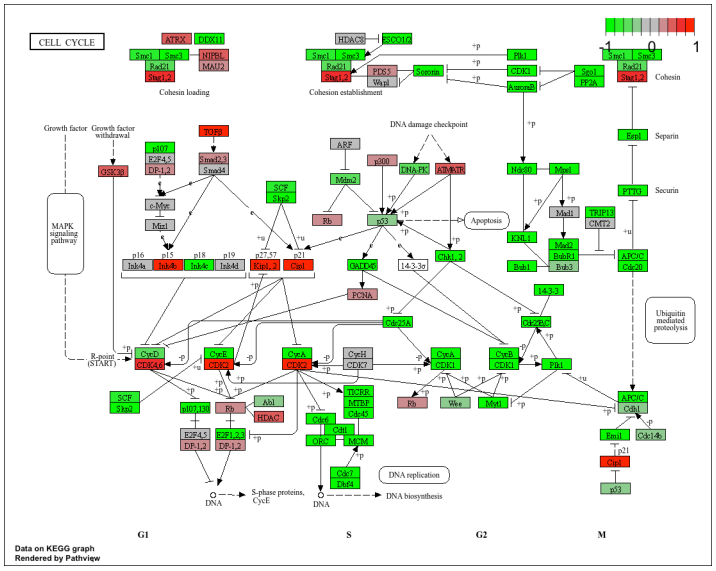
```r
files <- list.files(pattern = "pathview.png")
knitr::include_graphics(files)
```

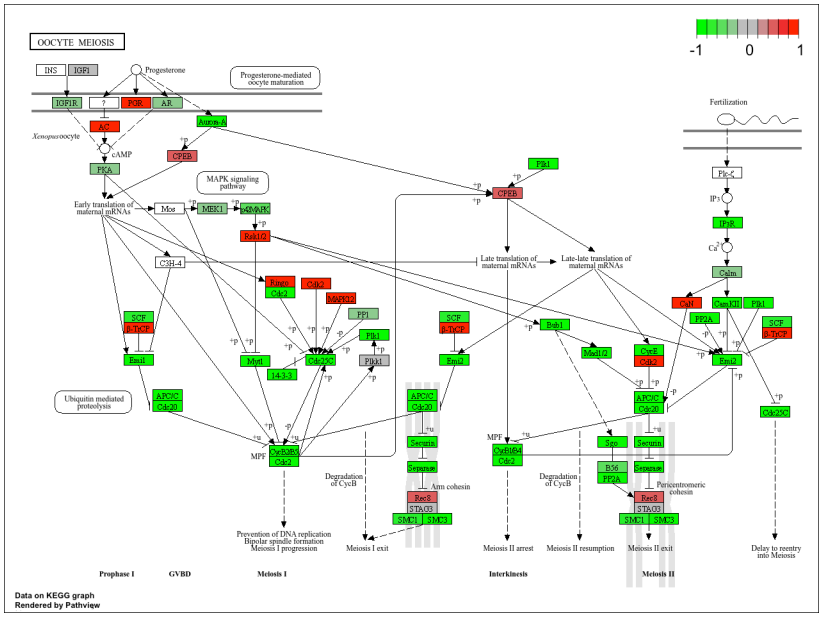


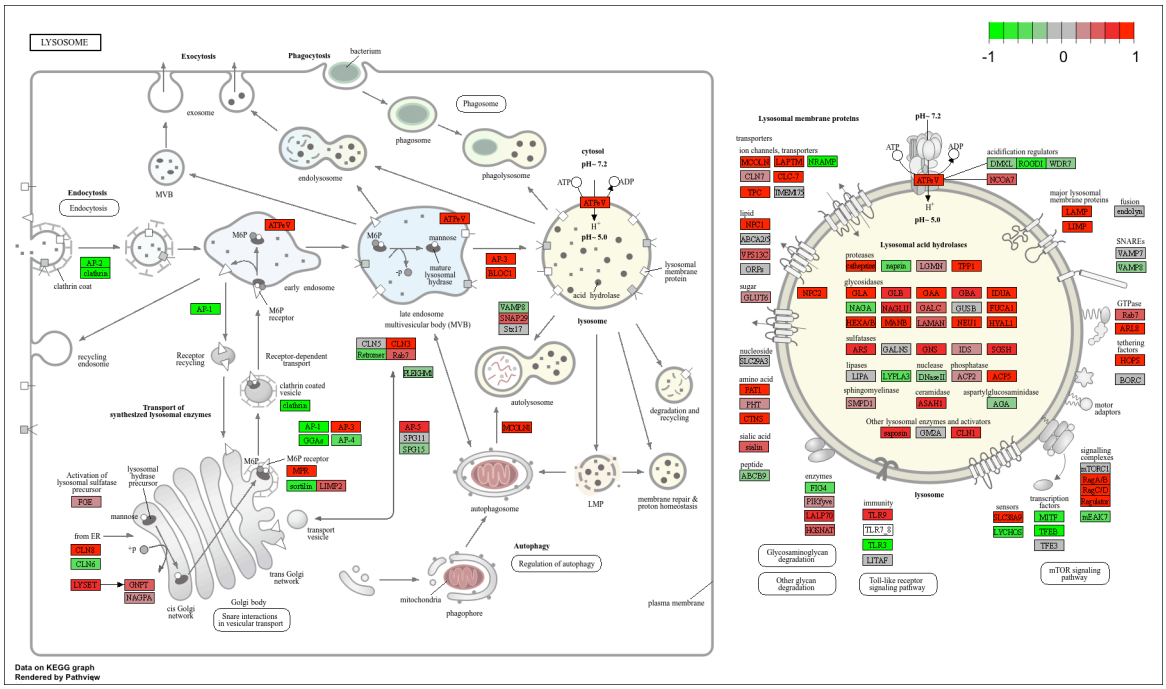

NUCLEOCYTOPLASMIC TRANSPORT
-1
0
1
Import
Export
mRNA Export
Importin
NLS
NPC
Cytoplasmic fibrils
Cytoplasm
Cytoplasmic ring
Lumen
Central channel
Spoke complex
Lumenal ring
Nucleoplasmic ring
Nucleus
Nuclear basket
Ran GTP
Exportin
NES
Pi
Rae1
DDX19
Nup358 complex
Nup214
Nup62 complex
Nup107-160 complex
ELYS
Nup153
Tpr
Upf1
Upf2
Tap
Ref/Aly
PYM
EJC
ARO
PABP
AAAAA
mRNA surveillance pathway
SRm160
Pinin
p15
CBC
m7G
Nuclear Pore complex (NPC)
Cytoplasmic fibrils
AL&DIM
hCG1
Gle1
DDX19
Rae1
Nup98
Nup214
Nup88
Nup358 complex
RanBP2
RanGAP1
UBC9
SUMO
Cytoplasmic ring / Nucleoplasmic ring (Symmetrical nups)
Nup160
Nup93
Seh13
Nup107
Nup133
Nup96
Nup45
Seh1
Nup43
Nup37
ELYS
Central channel
Nup62
Nup58/45
Nup54
Spoke complex
Nup205
Nup188
Nup155
Nup93
Nup53
Nup59
Lumenal ring
NDC1
Gp210
pom121
pom152
pom34
pom33
Nuclear basket
Tpr
Nup50
Nup153
Seoga2
Nup2
Nup1
Nup60
Nuclear transport complex
Importin
Adaptor proteins
IPOA
IPOB
SPN1
Exportin
XPO
Ran
eIF1A
PHAX
CBC
NMD3
Exon-junction complex (EJC)
EJC inner core
Y14
MAGOH
MLN51
EIF4A3
EJC outer shell
ACIN1
SAP18
RNPS1
Pinin
Ref/Aly
Transiently interacting factors
Upf1
Upf2
Upf3
Tap
p15
UAP56
SRm160
PYM
Transcription-export (TREX) complex
THO subcomplex
THOC1
THOC2
THOC5
THOC6
THOC7
TEX1
Data on KEGG graph
Rendered by Pathview

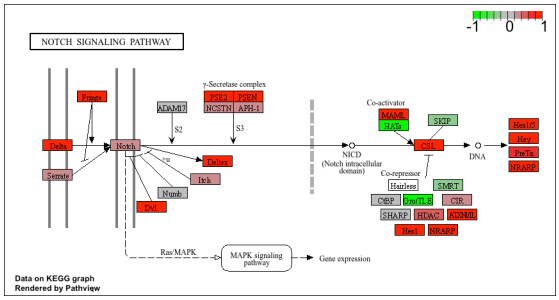

DNA REPLICATION
-1
0
1
Replication complex (Bacteria)
Lagging strand
Leading strand
RNase H / Pol I
Removal of RNA primer
Gap-filling
DNA ligase
Joining of Okazaki fragment
Pol III core
Clamp
Primase
DnaG
Primer
DnaB
SSB
Helicase
DNA polymerase III holoenzyme
DNA polymerase III holoenzyme
Pol III core
γδ complex
Clamp
Helicase
Primase
SSB
RNaseH
DNA polymerase I
DNA ligase
Replication complex (Archaea)
RNase H or Dna2
DNA ligase
DNA polymerase
Clamp loader
RFC
Primase
MCM
RPA/SSB
DNA polymerase B
DNA polymerase D
Helicase
Primase
RPA/SSB
Clamp
Clamp loader
RNaseH
Helicase
DNA ligase
Replication complex (Eukaryotes)
RNase H or Dna2
DNA ligase I
DNA polymerase δ complex
PCNA
RFC
DNA polymerase α-primase complex
α-Prim
MCM
RPA
Helicase
DNA polymerase ε complex
DNA polymerase α-primase complex
MCM complex (helicase)
RPA
Clamp
Clamp loader
RNaseH
RNaseHII
Helicase
DNA ligase
Data on KEGG graph
Rendered by Pathview

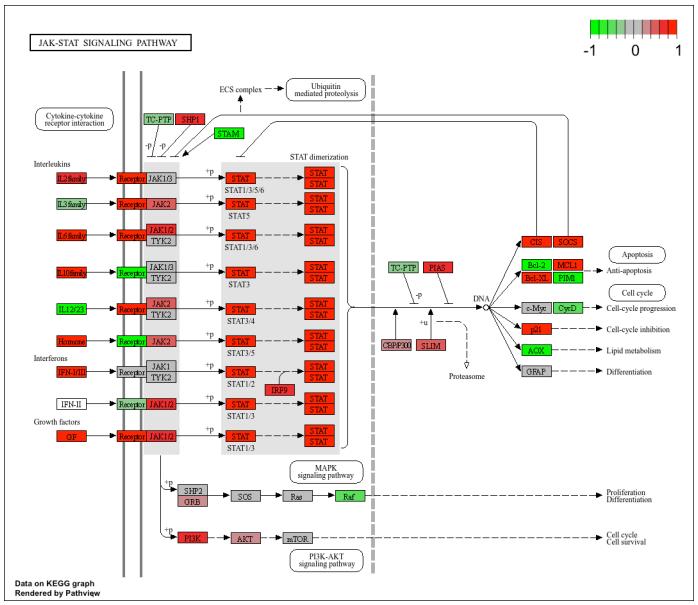

HOMOLOGOUS RECOMBINATION
Prokaryotic type
Eukaryotic type
Recognition
-1
0
1
5' to 3' resection
(Escherichia coli)
Double strand break
(Saccharomyces cerevisiae)
(Mammals)
RecJ
SSB
RecBCD
RecB
RecC
RecD
Filament formation
RecA
RecF
RecO
RecR
RecFOR
RecA filament
Strand invasion
DNA synthesis
DpoI
Holliday junction intermediate
Branch migration and resolution of Holliday junction
RuvA
RuvB
RuvC
RecG
RecFOR pathway
DpoIII
Replication restart
PriA
PriB
PriC
DnaT
RecBC pathway
Rad51B
Rad51C
Rad51D
XRCC2
XRCC3
Rad51 paralogs
MRX complex
MRN complex
Mre11
Rad50
Nbs1
XRS2
RPA
BRCA2-DSS1
Rad51
(S. cerevisiae)
(Common)
(Mammals)
Rad55
Rad57
Rad52
Rad51 filament
ATM
CtIP
BARD1
BRCA1
BRIP1
TOPBP1
Abraxas
RAP80
NBA1
BRE
BRCC36
PALB2
BRCA2
DSS1
SYCP3
Rad59
Rad54
D-loop
Second end capture
Branch migration and resolution of Holliday junction
BLM
TOP3
Mus81
Eme1
Crossover
or
Non-crossover
DSBR
Double-strand break repair
polB
Strand displacement
Flap removal and annealing
Ligation
SDSA
Synthesis-dependent strand annealing
BIR
Break-induced replication
Data on KEGG graph
Rendered by Pathview

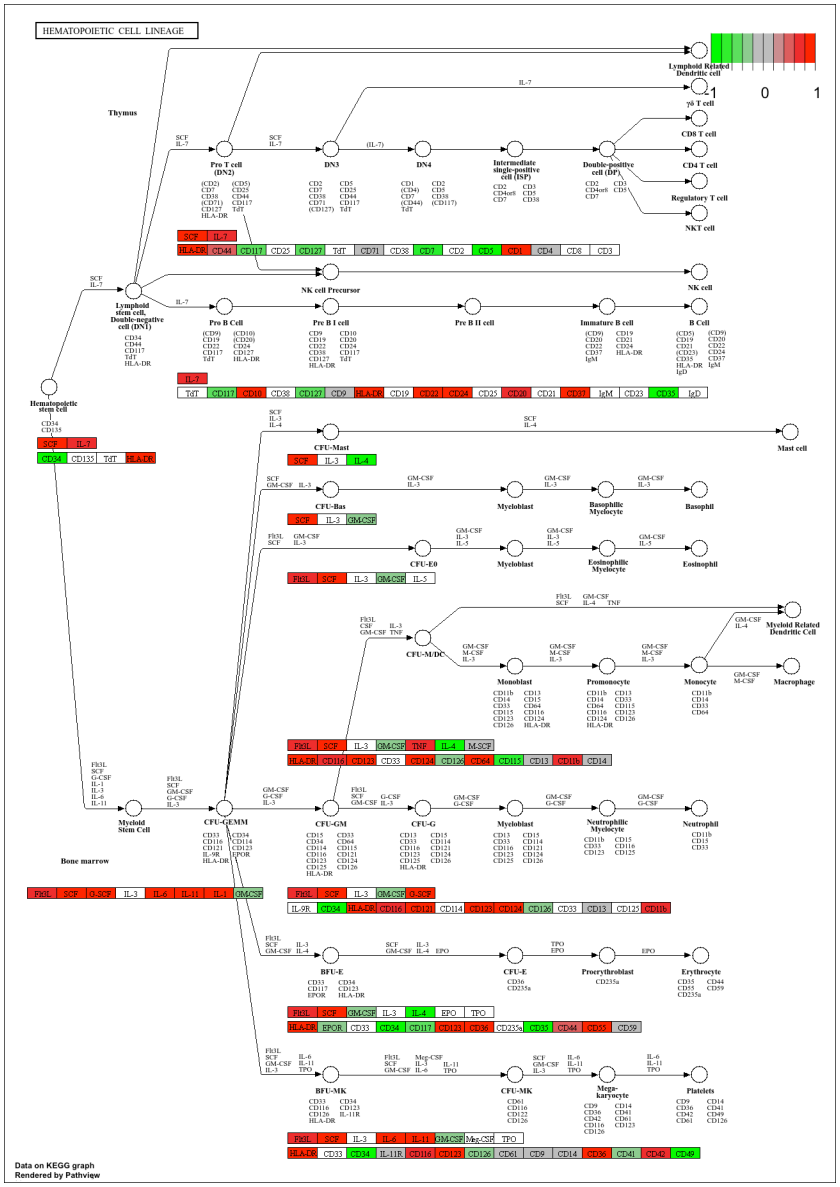

CELL CYCLE
-1
0
1
ATRX
DDX11
Smc1
Smc3
NIPBL
MAU2
Rad21
Stag1,2
Cohesin loading
HDAC8
ESCO1,2
Pds5
Wapl
Sororin
Cohesin establishment
Plk1
CDK1
AuroraB
Spo1
PP2A
Cohesin
Growth factor
Growth factor withdrawal
GSK3B
MAPK signaling pathway
p107
E2F4,5
DP1,2
TGFB
Smad2,3
Smad4
c-Myc
Mxi1
SCF
Skp2
ARF
Mdm2
p300
DNA-PK
ATM/ATR
DNA damage checkpoint
Hsp90
Myt1
Separin
PTTG1
Securin
Rb
p53
Apoptosis
Mad1
TRIP13
CMT2
CKI1,1
MAD2
Bub1
Bub1B
Bub3
APC/C
Cdc20
p16
p15
p18
p19
p27,57
Ink4a
Ink4b
Ink4c
Ink4d
Kip1,2
Cip1
p21
GADD45
14-3-3σ
CDK1,3
14-3-3
PCNA
Cdc25A
Cdc25C
Ubiquitin mediated proteolysis
R-point (START)
CycD
CDK4,6
CycE
CDK2
CycA
CDK2
CycH
CDK7
CycA
CDK1
CycB
CDK1
Plk1
TCFB
MTBP
Cdk4,6
Wee
Myt1
Abl
HDAC
Cdc6
Cdt1
Cdc7
Dbf4
DNA replication
DNA
S-phase proteins, CycE
DNA biosynthesis
Emi1
Cdc14b
p21
Cip1
p53
G1
S
G2
M
Data on KEGG graph
Rendered by Pathview


OOCYTE MEIOSIS
-1
0
1
Progesterone
Progesterone-mediated oocyte maturation
Fertilization
MAPK signaling pathway
Early translation of maternal mRNAs
Late translation of maternal mRNAs
Late-late translation of maternal mRNAs
Ubiquitin mediated proteolysis
MPF
Degradation of CycB
Arm cohesin
Pericentromeric cohesin
Prevention of DNA replication
Bipolar spindle formation
Meiosis I progression
Meiosis I exit
Meiosis II arrest
Meiosis II resumption
Meiosis II exit
Delay to reentry into Meiosis
Prophase I
GVBD
Meiosis I
Interkinesis
Meiosis II
Data on KEGG graph
Rendered by Pathview


LYSOSOME
-1
0
1
Exocytosis
Phagocytosis
bacterium
exosome
Phagosome
phagosome
endolysosome
phagolysosome
MVB
Endocytosis
clathrin coat
early endosome
late endosome
multivesicular body (MVB)
cytosol
pH~7.2
pH~5.0
lysosomal membrane protein
acid hydrolase
lysosome
recycling endosome
Receptor recycling
Receptor-dependent transport
clathrin coated vesicle
Transport of synthesized lysosomal enzymes
autolysosome
degradation and recycling
Activation of lysosomal sulfatase precursor
lysosomal hydrase precursor
mannose
from ER
trans Golgi network
cis Golgi network
Golgi body
Snare interactions in vesicular transport
transport vesicle
autophagosome
LMP
membrane repair & proton homeostasis
Autophagy
Regulation of autophagy
mitochondria
phagophore
plasma membrane
Lysosomal membrane proteins
transporters
ion channels, transporters
acidification regulators
major lysosomal membrane proteins
fusion
lipid
SNAREs
sugar
Lysosomal acid hydrolases
proteases
glycosidases
sulfatases
GTPase
tethering factors
nucleoside
lipases
nuclease
phosphatase
amino acid
sphingomyelinase
ceramidase
aspartylglucosaminidase
motor adaptors
sialic acid
Other lysosomal enzymes and activators
peptide
enzymes
immunity
signalling complexes
sensors
transcription factors
Glycosaminoglycan degradation
Other glycan degradation
Toll-like receptor signaling pathway
mTOR signaling pathway
Data on KEGG graph
Rendered by Pathview


NOTCH SIGNALING PATHWAY
-1 0 1
Fringe
γ-Secretase complex
PSE2
PSEN
NCSTN
APH-1
ADAM17
S2
S3
Co-activator
MAML
HATs
SKIP
Hes1/5
Hey
PreTa
NRARP
Delta
Notch
Serrate
Dulless
Itch
Numb
Dvl
NICD
(Notch intracellular domain)
CSL
DNA
Co-repressor
Hairless
SMRT
CtBP
Dsp/TLE
CIR
SHARP
HDAC1
ATP4B
Hes1
NRARP
Ras/MAPK
MAPK signaling pathway
Gene expression
Data on KEGG graph
Rendered by Pathview


JAK-STAT SIGNALING PATHWAY
-1 0 1
ECS complex
Ubiquitin mediated proteolysis
Cytokine-cytokine receptor interaction
TC-PTP
SHP1
STAM
STAT dimerization
Interleukins
IL2Rαβγ
IL3Rαβγ
IL6Rαβγ
IL10Rαβγ
IL12/23
Hormone
Receptor
JAK1/3
JAK2
JAK1/2
TYK2
JAK1/3
TYK2
JAK2
TYK2
JAK2
JAK1
TYK2
STAT
STAT1/3/5/6
STAT5
STAT1/3/6
STAT3
STAT3/4
STAT3/5
STAT1/2
IRF9
STAT1/3
STAT1/3
Interferons
IFN-I/III
IFN-II
Growth factors
GF
JAK1/2
JAK1/2
TC-PTP
PIAS
CBP/P300
SLIM
Proteasome
DNA
CIS
SOCS
Bcl-2
MCL1
Bcl-XL
FBM
c-Myc
CycD
p21
ACX
GFAP
Apoptosis
Anti-apoptosis
Cell cycle
Cell-cycle progression
Cell-cycle inhibition
Lipid metabolism
Differentiation
MAPK signaling pathway
SHP2
GRB
SOS
Ras
Raf
Proliferation
Differentiation
PI3K
AKT
mTOR
Cell cycle
Cell survival
PI3K-AKT signaling pathway
Data on KEGG graph
Rendered by Pathview



Q.Can you do the same procedure as above to plot the pathview figures for the top 5 down-regulated pathways?

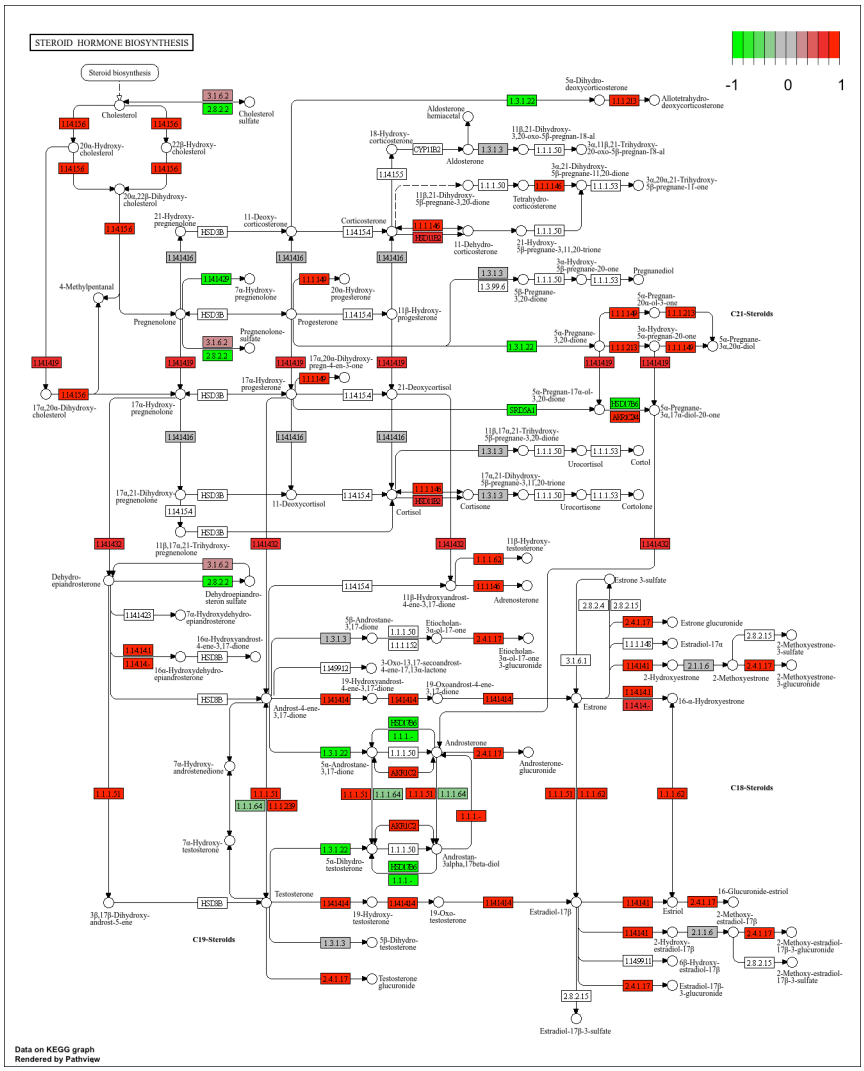
```
## Focus on top 5 downregulated pathways
keggrespathways_down <- rownames(keggres$less)[1:5]

## Extract KEGG IDs
keggresids_down <- substr(keggrespathways_down, start = 1, stop = 8)
keggresids_down
```

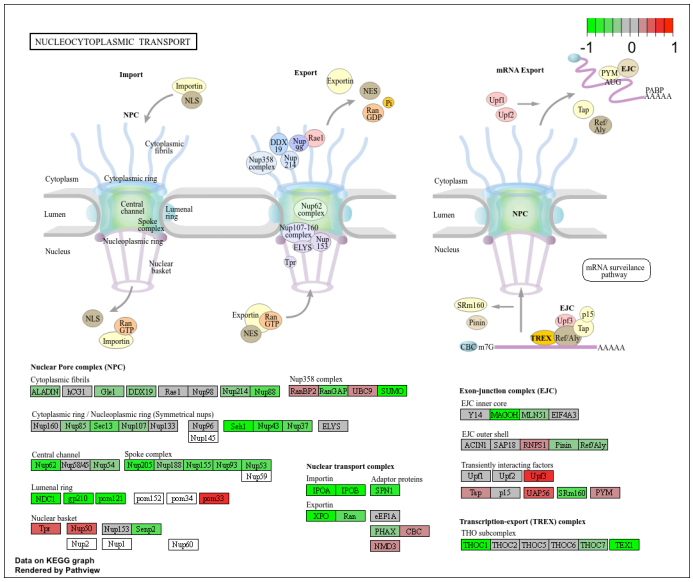
```
[1] "hsa04110" "hsa03030" "hsa03013" "hsa03440" "hsa04114"
```

```r
pathview(gene.data = foldchanges, pathway.id = keggresids_down, species = "hsa")
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/saharkamali/Desktop/BIMM 143 /Class 14/class14

Info: Writing image file hsa04110.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/saharkamali/Desktop/BIMM 143 /Class 14/class14

Info: Writing image file hsa03030.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/saharkamali/Desktop/BIMM 143 /Class 14/class14

Info: Writing image file hsa03013.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/saharkamali/Desktop/BIMM 143 /Class 14/class14

Info: Writing image file hsa03440.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/saharkamali/Desktop/BIMM 143 /Class 14/class14

Info: Writing image file hsa04114.pathview.png
```

```r
files_down <- list.files(pattern = "pathview.png")
knitr::include_graphics(files_down)
```

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-1
0
1
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NLS
NPC
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Cytoplasm
Cytoplasmic ring
Lumen
Central channel
Spoke complex
Lumenal ring
Nucleoplasmic ring
Nucleus
Nuclear basket
Ran GTP
Exportin
NES
Ran GDP
Rae1
DDX19
Nup358 complex
Nup214
Nup62 complex
Nup107-160 complex
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Tpr
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PYM
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AAAAA
Upf1
Upf2
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Ref/Aly
mRNA surveillance pathway
SRm160
Pinin
p15
TREX
CBC
m7G
Nuclear Pore complex (NPC)
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AladDH
hCG1
Gle1
DDX19
Rae1
Nup98
Nup214
Nup88
Nup358 complex
RanBP2
RanGAP1
UBC9
SUMO
Cytoplasmic ring / Nucleoplasmic ring (Symmetrical nups)
Nup160
Nup85
Seh13
Nup107
Nup133
Nup96
Nup45
Sec13
Nup43
Nup37
ELYS
Central channel
Nup62
Nup58/45
Nup54
Spoke complex
Nup205
Nup188
Nup155
Nup93
Nup53
Nup59
Lumenal ring
NDC1
Gp210
pom121
pom152
pom34
pom33
Nuclear basket
Tpr
Nup50
Nup153
Seoga2
Nup2
Nup1
Nup60
Nuclear transport complex
Importin
Adaptor proteins
IPOA
IPOB
SPN1
Exportin
XPO
Ran
eIF1A
PHAX
CBC
NMD3
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Y14
MAGOH
MLN51
EIF4A3
EJC outer shell
ACIN1
SAP18
RNPS1
Pinin
Ref/Aly
Transiently interacting factors
Upf1
Upf2
Upf3
Tap
p15
UAP56
SRm160
PYM
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THOC6
THOC7
TEX1
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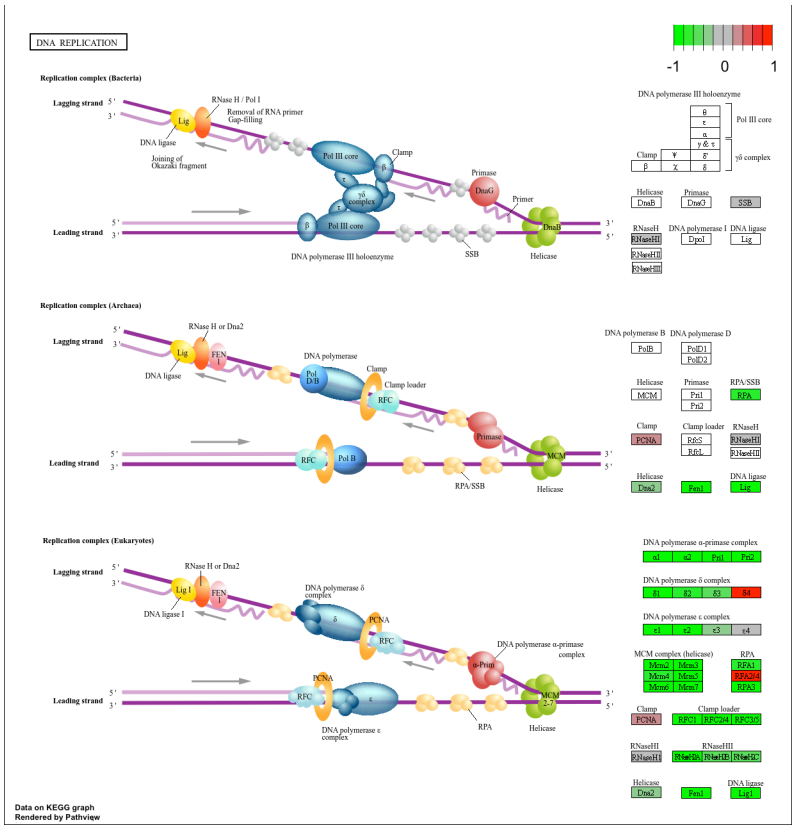
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Leading strand
RNase H / Pol I
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Gap-filling
DNA ligase
Joining of Okazaki fragment
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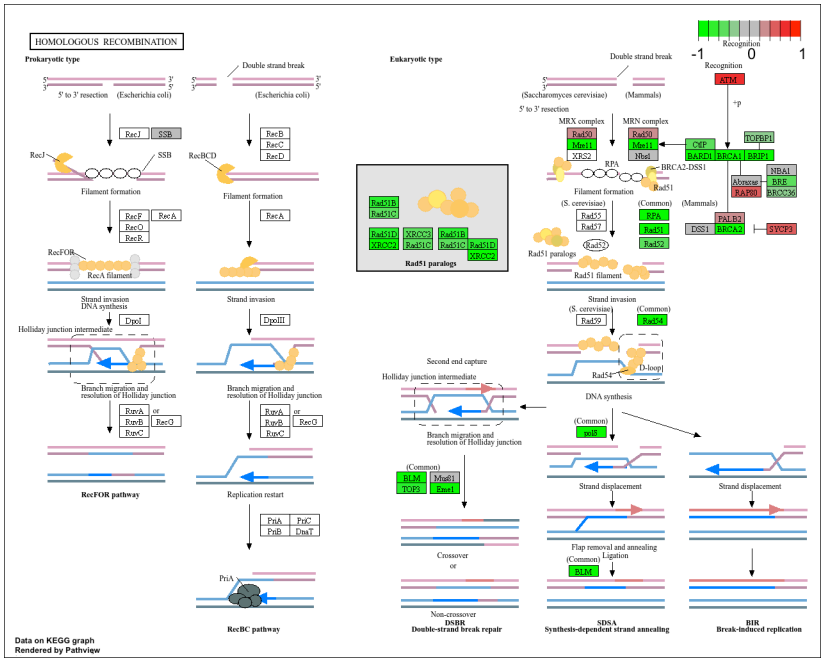
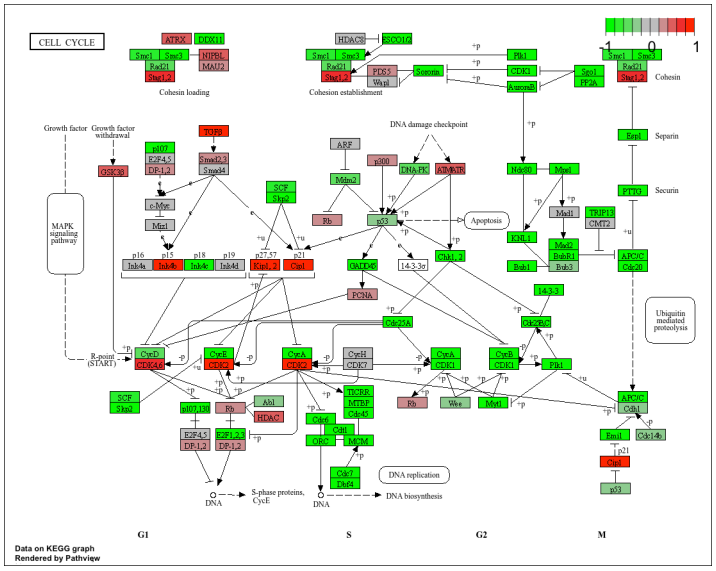
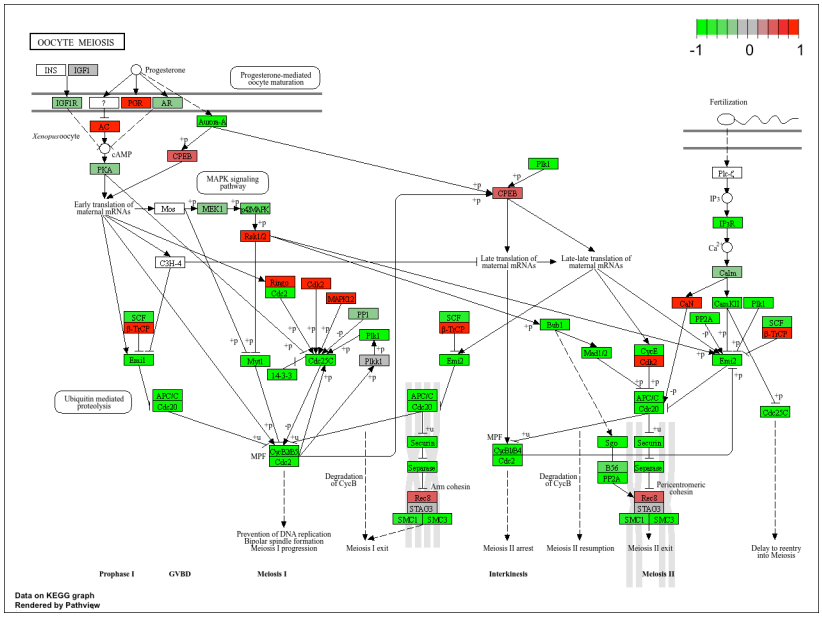
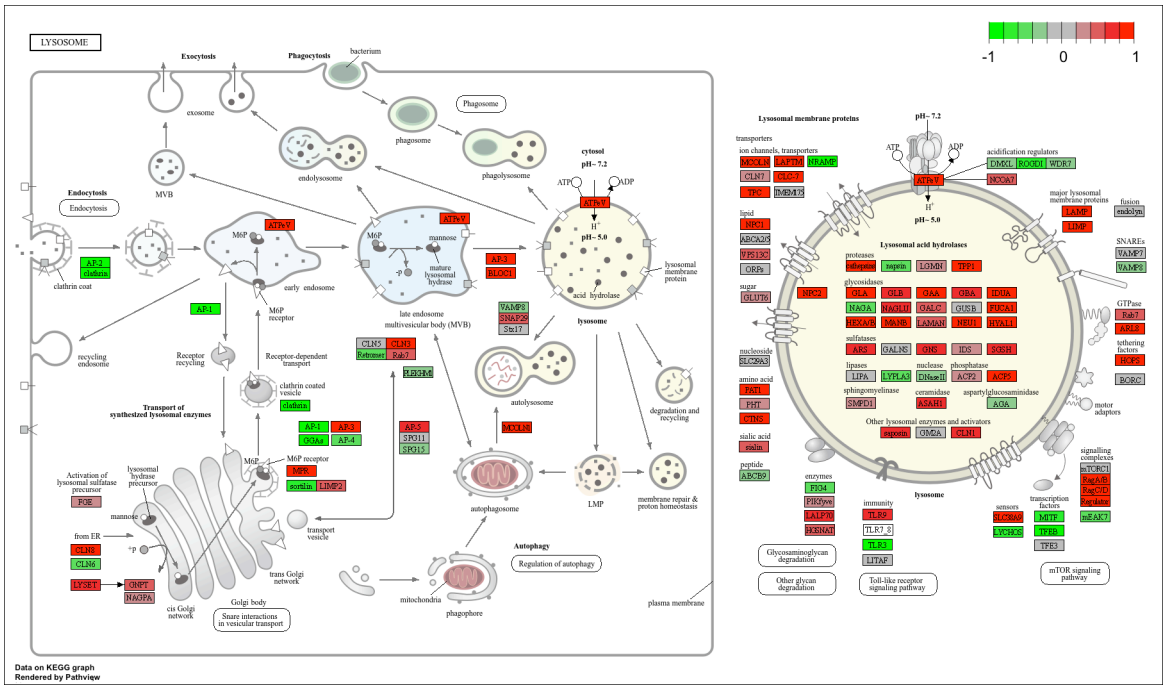
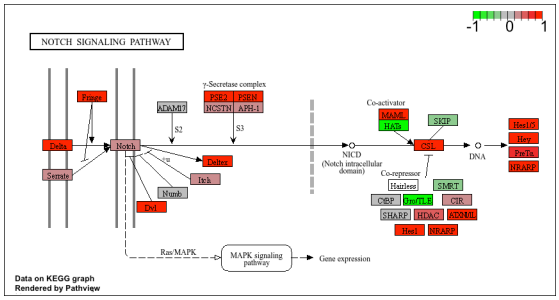
Clamp
Primase
DnaG
Primer
DnaB
SSB
Helicase
DNA polymerase III holoenzyme
DNA polymerase III holoenzyme
Pol III core
γδ complex
Clamp
Helicase
Primase
SSB
RNaseH
DNA polymerase I
DNA ligase
Replication complex (Archaea)
RNase H or Dna2
DNA ligase
DNA polymerase
Clamp loader
RFC
Primase
MCM
RPA/SSB
DNA polymerase B
DNA polymerase D
Helicase
Primase
RPA/SSB
Clamp
Clamp loader
RNaseH
Helicase
DNA ligase
Replication complex (Eukaryotes)
RNase H or Dna2
DNA ligase I
DNA polymerase δ complex
PCNA
RFC
DNA polymerase α-primase complex
α-Prim
MCM
RPA
Helicase
DNA polymerase ε complex
DNA polymerase α-primase complex
DNA polymerase δ complex
DNA polymerase ε complex
MCM complex (helicase)
RPA
Clamp
Clamp loader
RNaseH
RNaseHII
Helicase
DNA ligase
Data on KEGG graph
Rendered by Pathview

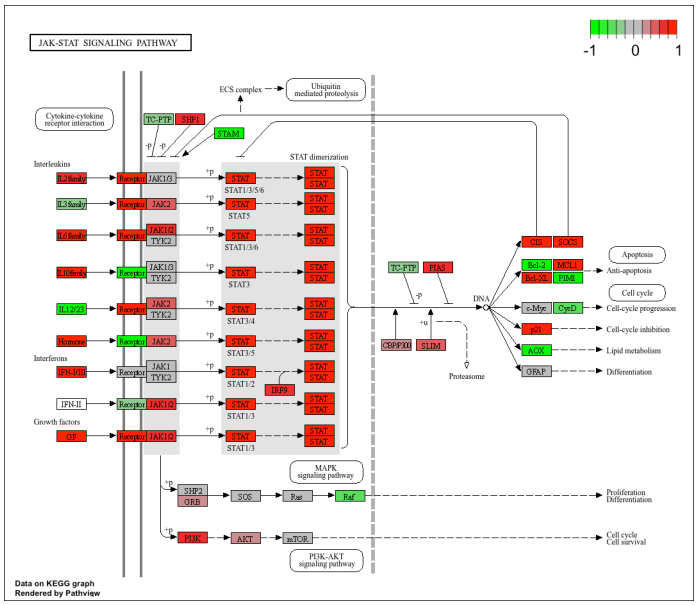

HOMOLOGOUS RECOMBINATION
Prokaryotic type
Eukaryotic type
Recognition
-1
0
1
5' to 3' resection
(Escherichia coli)
Double strand break
RecJ
SSB
RecBCD
RecB
RecC
RecD
Filament formation
RecF
RecA
RecO
RecR
RecFOR
RecA filament
Strand invasion
DNA synthesis
DpoI
Holliday junction intermediate
Branch migration and resolution of Holliday junction
RuvA
RuvB
RuvC
or
RecG
RecFOR pathway
DpoIII
Replication restart
PriA
PriB
PriC
DnaT
RecBC pathway
Rad51B
Rad51C
Rad51D
XRCC2
XRCC3
Rad51 paralogs
(Saccharomyces cerevisiae)
(Mammals)
MRX complex
Mre11
Rad50
XRS2
MRN complex
Nbs1
ATM
CtIP
BARD1
BRCA1
BRIP1
TOPBP1
RPA
BRCA2-DSS1
Rad51
Abraxas
RAP80
NBA1
BRE
BRCC36
PALB2
DSS1
SYCP3
(S. cerevisiae)
(Common)
Rad55
Rad57
Rad52
Rad51 filament
Rad59
Rad54
D-loop
Second end capture
BLM
TOP3
Mus81
Eme1
Crossover
Non-crossover
DSBR
Double-strand break repair
Strand displacement
Flap removal and annealing
Ligation
SDSA
Synthesis-dependent strand annealing
BIR
Break-induced replication
Data on KEGG graph
Rendered by Pathview

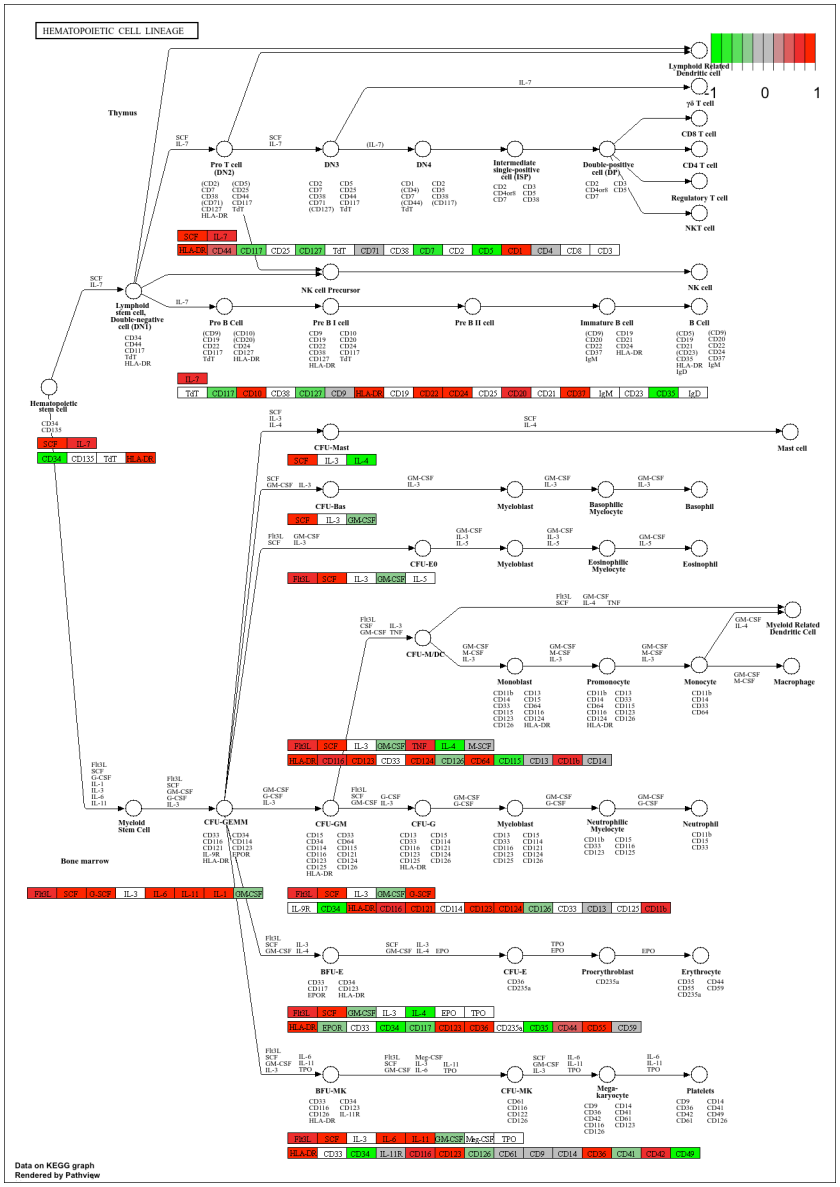

CELL CYCLE
-1
0
1
ATRX
DDX11
Smc1
Smc3
NIPBL
MAU2
Rad21
Stag1,2
Cohesin loading
HDAC8
ESCO1,2
Pds5
Wapl
Sororin
Cohesin establishment
Plk1
CDK1
AuroraB
Sgo1
PP2A
Cohesin
Growth factor
Growth factor withdrawal
GSK3B
TGFβ
p107
E2F4,5
DP1,2
Smad2,3
Smad4
c-Myc
Mxi1
MAPK signaling pathway
DNA damage checkpoint
ARF
Mdm2
p300
DNA-PK
ATM/ATR
SCF
Skp2
Rb
p53
Apoptosis
Hec1/2
Myt1
Emi1
Separin
PTTG1
Securin
Mad1
TRIP13
CMT2
KIM-1
MAD2
Bub1
Bub3
APC/C
Cdc20
p16
p15
p18
p19
p27,57
Ink4a
Ink4b
Ink4c
Ink4d
Kip1,2
Cip1
p21
GADD45
14-3-3σ
CDK1,3
PCNA
Cdc25A
Cdc25C
Ubiquitin mediated proteolysis
R-point (START)
CycD
CDK4,6
CycE
CDK2
CycA
CDK2
CycH
CDK7
CycA
CDK1
CycB
CDK1
Plk1
TCFB
MTBP
Cdk-M1
Wee
Abl
HDAC
Cdc6
Cdt1
ORC
MCM
Cdk7
Hrd1
DNA replication
DNA
S-phase proteins, CycE
DNA biosynthesis
Cdc14b
Cdk1-4b
p21
Cip1
p53
G1
S
G2
M
Data on KEGG graph
Rendered by Pathview


OOCYTE MEIOSIS
-1
0
1
Progesterone
Progesterone-mediated oocyte maturation
Fertilization
MAPK signaling pathway
Late translation of maternal mRNAs
Late-late translation of maternal mRNAs
Ubiquitin mediated proteolysis
Degradation of CycB
Arm cohesin
Pericentromeric cohesin
Prevention of DNA replication
Bipolar spindle formation
Meiosis I progression
Meiosis I exit
Meiosis II arrest
Meiosis II resumption
Meiosis II exit
Delay to reentry into Meiosis
Prophase I
GVBD
Meiosis I
Interkinesis
Meiosis II
Data on KEGG graph
Rendered by Pathview


LYSOSOME
-1
0
1
Exocytosis
Phagocytosis
bacterium
exosome
Phagosome
phagosome
endolysosome
phagolysosome
MVB
Endocytosis
clathrin coat
early endosome
late endosome
multivesicular body (MVB)
cytosol
pH~7.2
pH~5.0
lysosomal membrane protein
acid hydrolase
lysosome
recycling endosome
Receptor recycling
Receptor-dependent transport
clathrin coated vesicle
Transport of synthesized lysosomal enzymes
autolysosome
degradation and recycling
Activation of lysosomal sulfatase precursor
lysosomal hydrase precursor
mannose
from ER
trans Golgi network
cis Golgi network
Golgi body
Snare interactions in vesicular transport
transport vesicle
autophagosome
LMP
membrane repair & proton homeostasis
Autophagy
Regulation of autophagy
mitochondria
phagophore
plasma membrane
Lysosomal membrane proteins
transporters
ion channels, transporters
acidification regulators
major lysosomal membrane proteins
fusion
lipid
SNAREs
sugar
Lysosomal acid hydrolases
proteases
glycosidases
sulfatases
lipases
nuclease
phosphatase
sphingomyelinase
ceramidase
aspartylglucosaminidase
Other lysosomal enzymes and activators
GTPase
tethering factors
nucleoside
amino acid
sialic acid
peptide
motor adaptors
signalling complexes
enzymes
immunity
sensors
transcription factors
Glycosaminoglycan degradation
Other glycan degradation
Toll-like receptor signaling pathway
mTOR signaling pathway
Data on KEGG graph
Rendered by Pathview


-1 0 1
NOTCH SIGNALING PATHWAY
Fringe
Delta
Serrate
Notch
ADAM17
S2
γ-Secretase complex
PSE2
PSEN
NCSTN
APH-1
S3
Dulless
Itch
Numb
Dvl
NICD
(Notch intracellular domain)
Co-activator
MAML
HATs
SKIP
CSL
DNA
Co-repressor
Hairless
SMRT
CtBP
Dsp/TLE
CIR
SHARP
HDAC1
ATP4B
Hes1
NRARP
Hes1/5
Hey
PreTa
NRARP
Ras/MAPK
MAPK signaling pathway
Gene expression
Data on KEGG graph
Rendered by Pathview


-1 0 1
JAK-STAT SIGNALING PATHWAY
ECS complex
Ubiquitin mediated proteolysis
Cytokine-cytokine receptor interaction
TC-PTP
SHP1
STAM
STAT dimerization
Interleukins
IL2family
IL3family
IL6family
IL10family
IL12/23
Hormone
Interferons
IFN-I/III
IFN-II
Growth factors
GF
Receptor
JAK1/3
JAK2
JAK1/2
TYK2
JAK1
STAT
STAT1/3/5/6
STAT5
STAT1/3/6
STAT3
STAT3/4
STAT3/5
STAT1/2
IRF9
STAT1/3
TC-PTP
PIAS
CIS
SOCS
Bcl-2
MCL1
Bcl-XL
FIMS
c-Myc
CycD
p21
ACX
GFAP
DNA
CEBPA/00
SLIM
Proteasome
Apoptosis
Anti-apoptosis
Cell cycle
Cell-cycle progression
Cell-cycle inhibition
Lipid metabolism
Differentiation
MAPK signaling pathway
SHP2
GRB
SOS
Ras
Raf
Proliferation
Differentiation
PI3K
AKT
mTOR
Cell cycle
Cell survival
PI3K-AKT signaling pathway
Data on KEGG graph
Rendered by Pathview



##Section 3. Gene Ontology (GO)

```
data(go.sets.hs)
data(go.subs.hs)

# Focus on Biological Process subset of GO
gobpsets = go.sets.hs[go.subs.hs$BP]

gobpres = gage(foldchanges, gsets=gobpsets)
```

```r
lapply(gobpres, head)
```

```
$greater
                                          p.geomean stat.mean        p.val
GO:0007156 homophilic cell adhesion       8.132355e-05  3.836423 8.132355e-05
GO:0002009 morphogenesis of an epithelium 1.301491e-04  3.672346 1.301491e-04
GO:0048729 tissue morphogenesis           1.324453e-04  3.663702 1.324453e-04
GO:0007610 behavior                       1.786592e-04  3.585306 1.786592e-04
GO:0035295 tube development               5.558494e-04  3.273391 5.558494e-04
GO:0060562 epithelial tube morphogenesis  5.613074e-04  3.277405 5.613074e-04
                                              q.val set.size         exp1
GO:0007156 homophilic cell adhesion       0.1804788      113 8.132355e-05
GO:0002009 morphogenesis of an epithelium 0.1804788      339 1.301491e-04
GO:0048729 tissue morphogenesis           0.1804788      424 1.324453e-04
GO:0007610 behavior                       0.1825897      426 1.786592e-04
GO:0035295 tube development               0.3325708      391 5.558494e-04
GO:0060562 epithelial tube morphogenesis  0.3325708      257 5.613074e-04

$less
                                            p.geomean stat.mean        p.val
GO:0048285 organelle fission             1.735928e-15 -8.047667 1.735928e-15
GO:0000280 nuclear division              4.808128e-15 -7.923688 4.808128e-15
GO:0007067 mitosis                       4.808128e-15 -7.923688 4.808128e-15
GO:0000087 M phase of mitotic cell cycle 1.312717e-14 -7.781713 1.312717e-14
GO:0007059 chromosome segregation        2.149699e-11 -6.868718 2.149699e-11
GO:0000236 mitotic prometaphase          1.794819e-10 -6.689146 1.794819e-10
                                                q.val set.size         exp1
GO:0048285 organelle fission             6.551876e-12      376 1.735928e-15
GO:0000280 nuclear division              6.551876e-12      352 4.808128e-15
GO:0007067 mitosis                       6.551876e-12      352 4.808128e-15
GO:0000087 M phase of mitotic cell cycle 1.341596e-11      362 1.312717e-14
GO:0007059 chromosome segregation        1.757594e-08      142 2.149699e-11
GO:0000236 mitotic prometaphase          1.222870e-07       84 1.794819e-10

$stats
                                          stat.mean     exp1
GO:0007156 homophilic cell adhesion        3.836423 3.836423
GO:0002009 morphogenesis of an epithelium  3.672346 3.672346
GO:0048729 tissue morphogenesis            3.663702 3.663702
GO:0007610 behavior                        3.585306 3.585306
GO:0035295 tube development                3.273391 3.273391
```

```
GO:0060562 epithelial tube morphogenesis   3.277405 3.277405
```

##Section 4. Reactome Analysis

```r
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8147"
```

```r
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

> Q: What pathway has the most significant "Entities p-value"? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

what pathway has the most significant entities p-value? Cell Cycle, Mitotic entities p-value: 2.1×10

Do these match our KEGG results? It's pretty consistent with the KEGG results since they both point to cell cycle/mitosis. Any differences are mostly because Reactome and KEGG group pathways differently, and the tests aren't the same (Reactome uses a "significant genes" list; GAGE uses the fold-change values for all genes).